Fake News Detection Using BERT

Project Description Report

UML501 Machine Learning

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1. Introduction

With the rapid growth of digital media and online news platforms, the spread of fake news has become a serious global issue. Fake news refers to false or misleading information presented as legitimate news with the intent to deceive readers. Such misinformation can influence public opinion, create social unrest, and affect political and economic decisions. Manual verification of news articles is time-consuming and impractical at large scales, making automated fake news detection an important research problem.

Recent advancements in Natural Language Processing (NLP) and deep learning have enabled machines to understand textual data more effectively. Transformer-based models, especially BERT (Bidirectional Encoder Representations from Transformers), have achieved state-of-the-art performance in many NLP tasks due to their ability to capture contextual relationships between words.

In this project, a fake news detection system is developed using the BERT model to classify news articles as **Fake** or **Real**. The model is trained and evaluated using a publicly available dataset, and its performance is analysed using multiple evaluation metrics and visualizations.

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2. Objectives

The main objectives of this project are:

1. To understand the problem of fake news detection and its challenges.
2. To build a binary text classification system using deep learning.
3. To fine-tune a pre-trained BERT model for fake news classification.
4. To evaluate the model using standard machine learning metrics.
5. To analyze and visualize the dataset and model predictions.
6. To demonstrate the effectiveness of transformer-based NLP models.

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3. Dataset Used

3.1 Dataset Description

The dataset used in this project consists of two CSV files:

- **Fake.csv** – Contains fake news articles
- **True.csv** – Contains real news articles

Each file includes textual information such as the news title or article content. A binary label is assigned to each article:

- **0 → Fake News**
- **1 → Real News**

The two datasets are merged into a single dataset and shuffled before training.

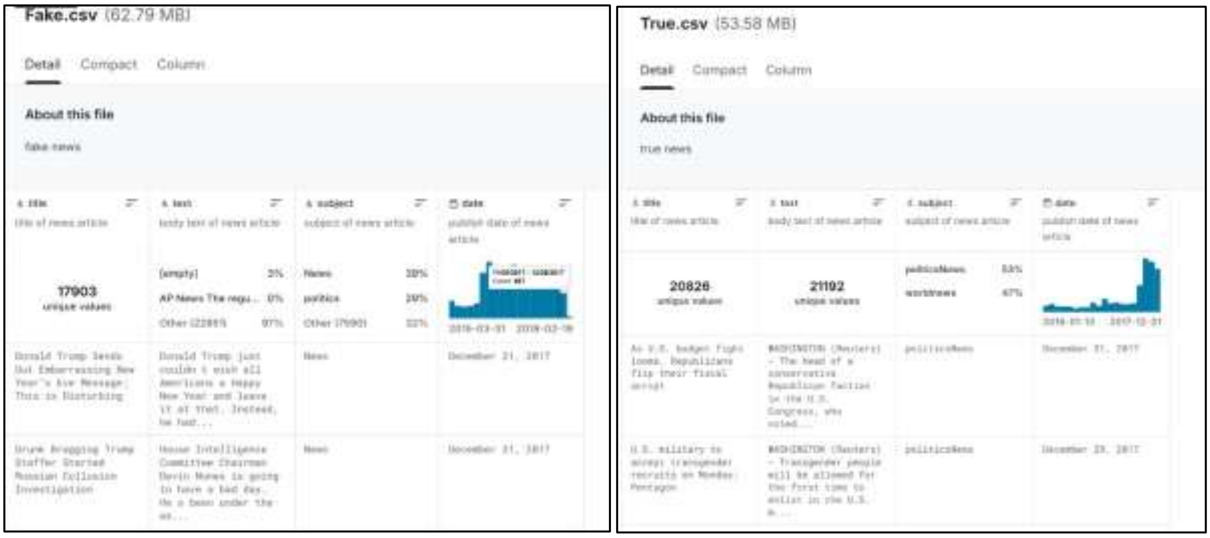
3.2 Dataset Download Link

The dataset was obtained from Kaggle and can be downloaded from the following link:
https://www.kaggle.com/datasets/clmentbisaillon/fake-and-real-news-dataset

3.3 Dataset Statistics

- Total samples: Approximately 44,000 news articles
- Classes: Fake and Real
- Data split:
  - Training set: 64%
  - Validation set: 16%
  - Test set: 20%

Minimal preprocessing is applied, such as removing extra whitespaces. Advanced preprocessing techniques like stemming or lemmatization are avoided since BERT is designed to work with raw text and contextual information.



Screenshot of the dataset files (Fake.csv and True.csv) or a snapshot showing dataset loading and shape output.

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4. Methodology

4.1 Model Used

This project implements the **BERT (bert-base-uncased)** model, which is a transformer-based pre-trained language model. BERT reads text in both forward and backward directions, allowing it to understand context more effectively than traditional models.
Only BERT was implemented in this project due to its superior performance compared to traditional machine learning and recurrent neural network models for text classification tasks.

4.2 Tokenization

The BERT tokenizer converts text into tokens using WordPiece tokenization. Each input text is:

- Truncated or padded to a fixed maximum length
- Converted into input IDs and attention masks
- Prepared in a format suitable for the BERT model

4.3 Training Process

The model is fine-tuned using:

- **Optimizer:** AdamW
- **Loss Function:** Cross-Entropy Loss
- **Learning Rate Scheduler:** Linear scheduler with warm-up
- **Number of Epochs:** 3

During training, the model parameters are updated using backpropagation. Validation is performed after each epoch to monitor model performance.

```
•••  Epoch 1: Train loss=0.1277, acc=0.9508
     Val   : loss=0.0496, acc=0.9825
     Epoch 2: Train loss=0.0253, acc=0.9913
     Val   : loss=0.0397, acc=0.9861
     Epoch 3: Train loss=0.0069, acc=0.9978
     Val   : loss=0.0455, acc=0.9887
```

Screenshot showing epoch-wise training and validation loss/accuracy output.

5. Evaluation Metrics Used

To evaluate the performance of the fake news detection model, the following metrics are used:

1. **Accuracy** – Measures the overall correctness of predictions.
2. **Precision** – Measures how many predicted real news articles are actually real.
3. **Recall** – Measures how many real news articles are correctly identified.
4. **F1-Score** – Harmonic mean of precision and recall.
5. **Confusion Matrix** – Displays correct and incorrect classifications.
6. **ROC Curve and AUC Score** – Evaluates performance across different thresholds.
7. **Precision–Recall Curve** – Useful for analyzing class-wise performance.

6. Experimental Results

After training and testing the BERT model on the dataset, strong classification performance was achieved.

6.1 Quantitative Results

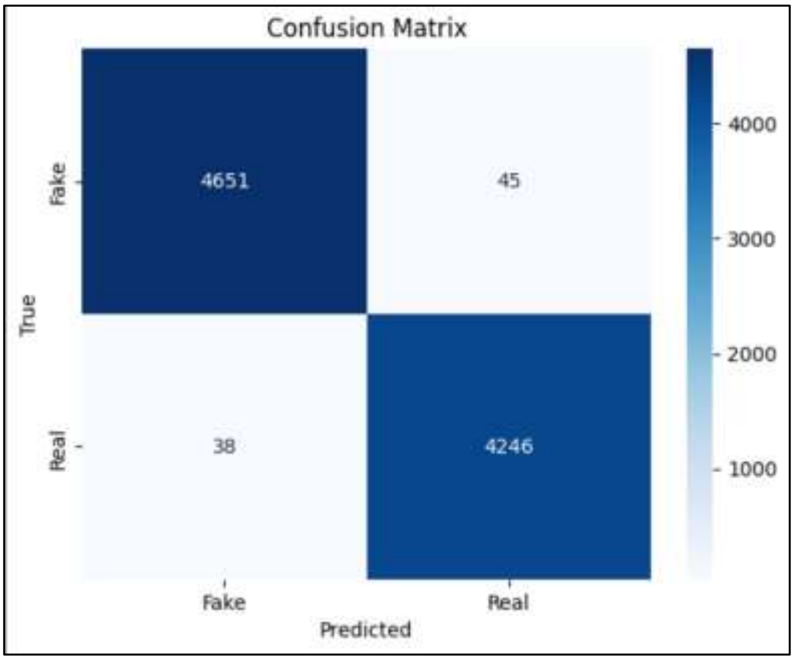
- High test accuracy (above 90%)
- High precision, recall, and F1-score for both classes
- Low number of misclassifications

```
Test accuracy: 0.9907572383073496

Classification report:

              precision    recall  f1-score   support

           0     0.9919    0.9904    0.9912      4696
           1     0.9895    0.9911    0.9903      4284

    accuracy                         0.9908      8980
   macro avg     0.9907    0.9908    0.9907      8980
weighted avg     0.9908    0.9908    0.9908      8980
```

Screenshot of the classification report showing precision, recall, F1-score, and accuracy.

6.2 Confusion Matrix Analysis

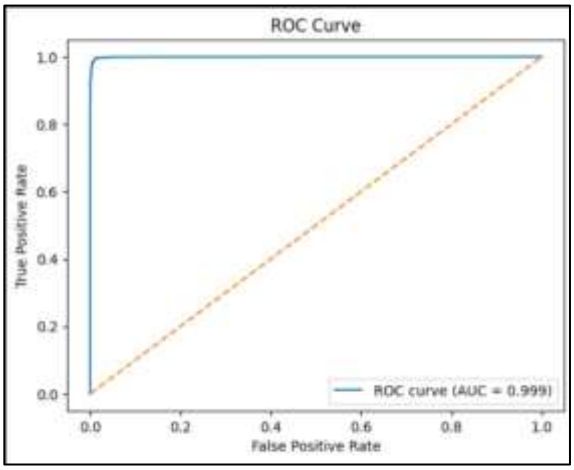
The confusion matrix shows that most fake and real news articles are correctly classified. Very few fake articles are misclassified as real and vice versa, indicating robust model performance.



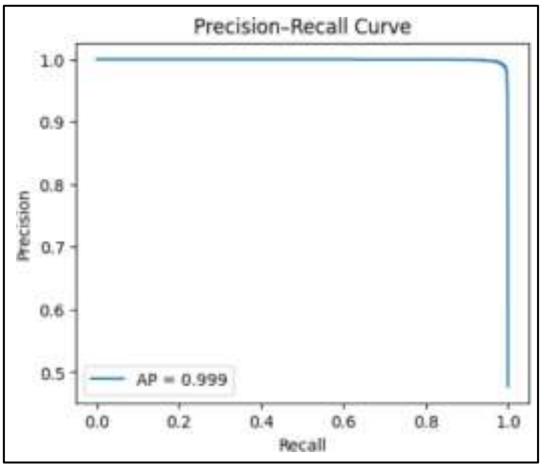
Screenshot of the confusion matrix.

6.3 ROC and Precision–Recall Analysis

The ROC curve shows a high Area Under the Curve (AUC), indicating strong separability between fake and real news. The precision–recall curve further confirms balanced performance across both classes.



Screenshot of the ROC curve



Screenshot of the Precision–Recall curve

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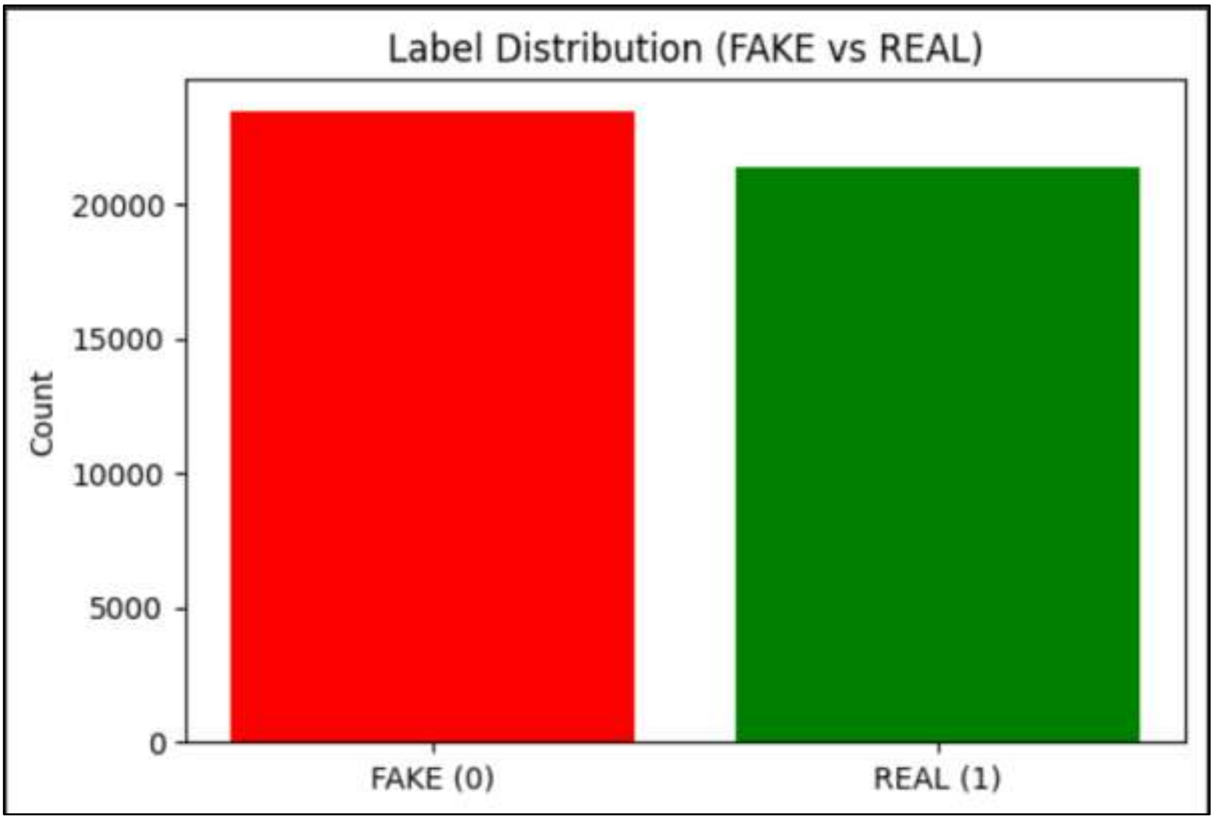
7. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand the dataset characteristics.

7.1 Label Distribution

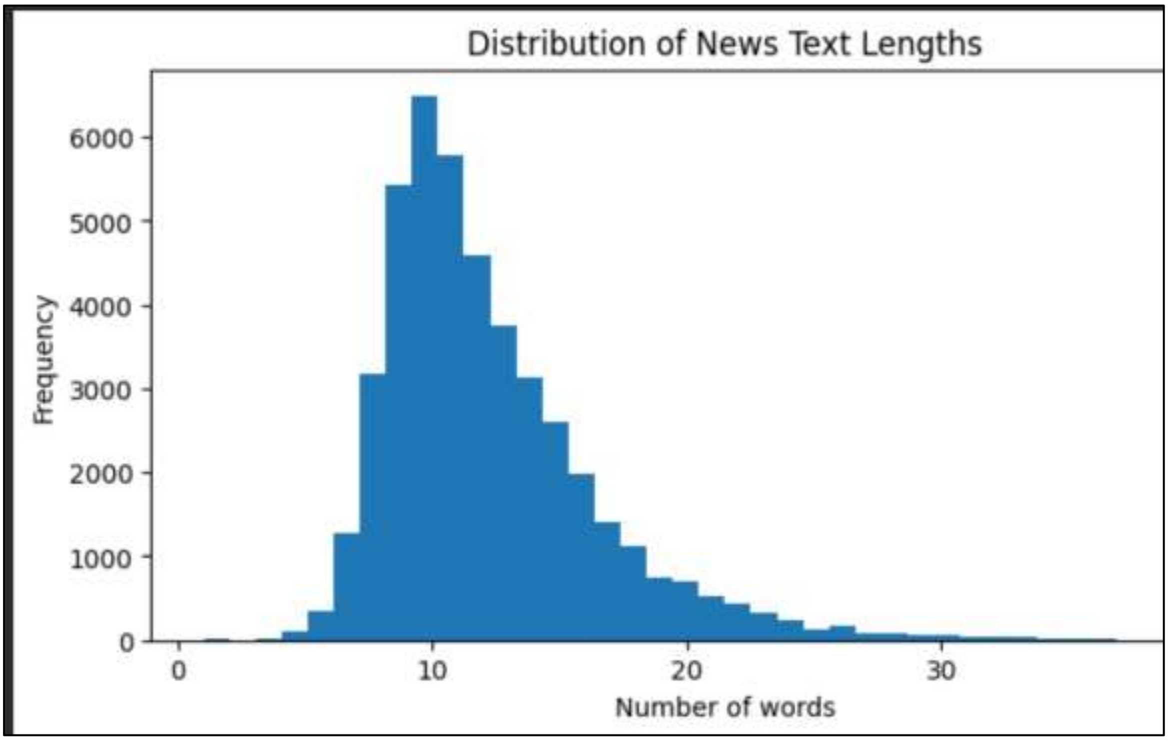
The dataset contains a balanced number of fake and real news articles.



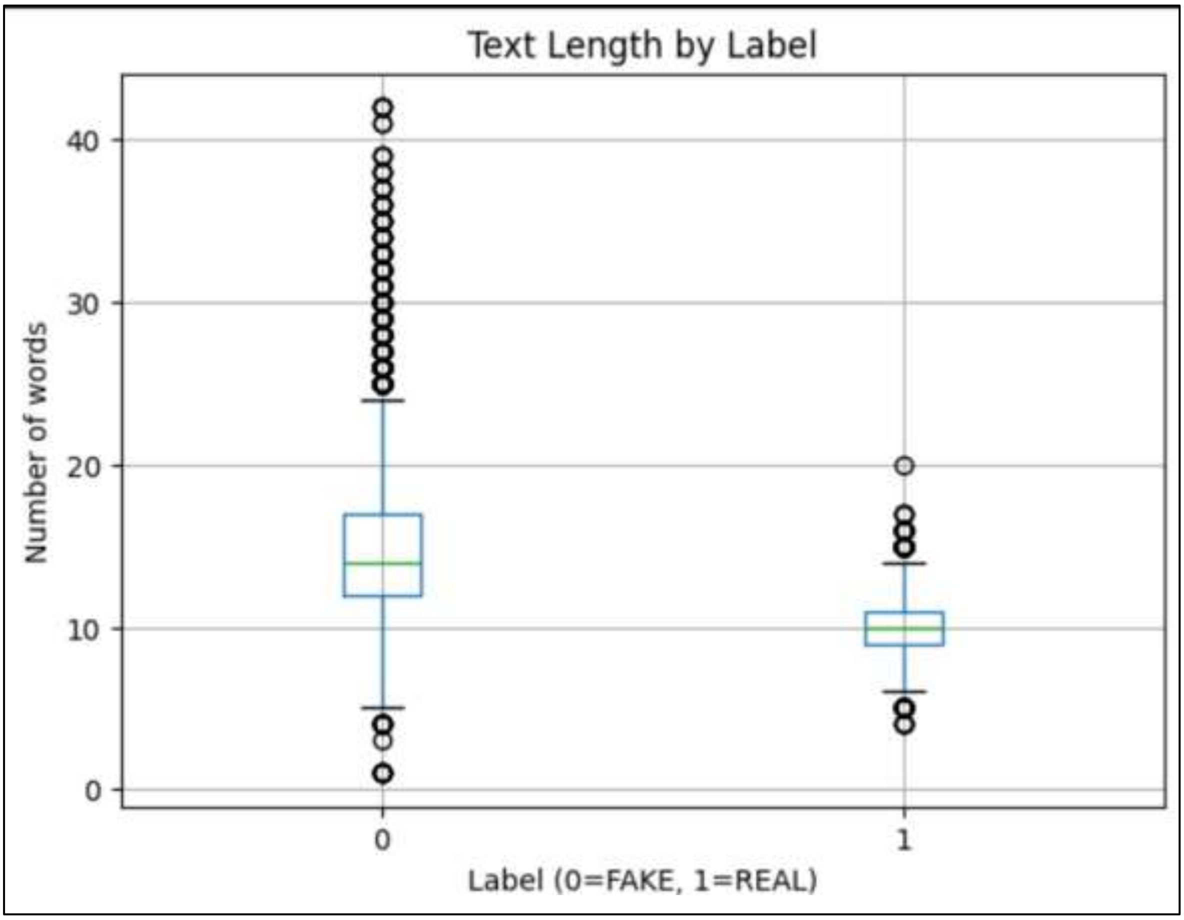
Screenshot of the label distribution bar chart.

7.2 Text Length Analysis
The distribution of text lengths shows variation between fake and real news articles, which helps the model learn different writing patterns.

- *Histogram of text length distribution*



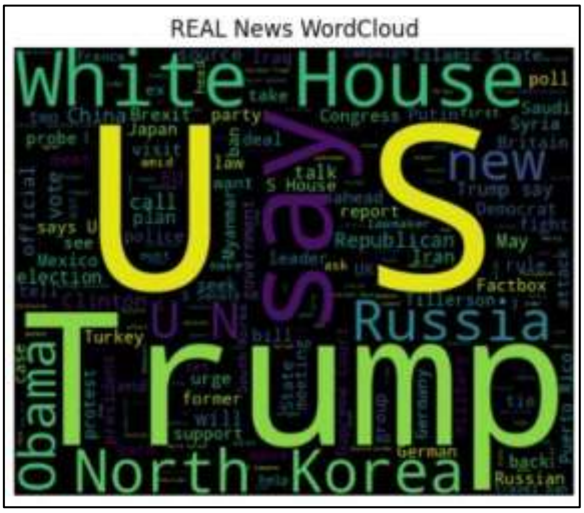
- *Box plot comparing text length by label*



7.3 Word Cloud Analysis

Word clouds reveal commonly used words in fake and real news articles, highlighting stylistic and vocabulary differences.

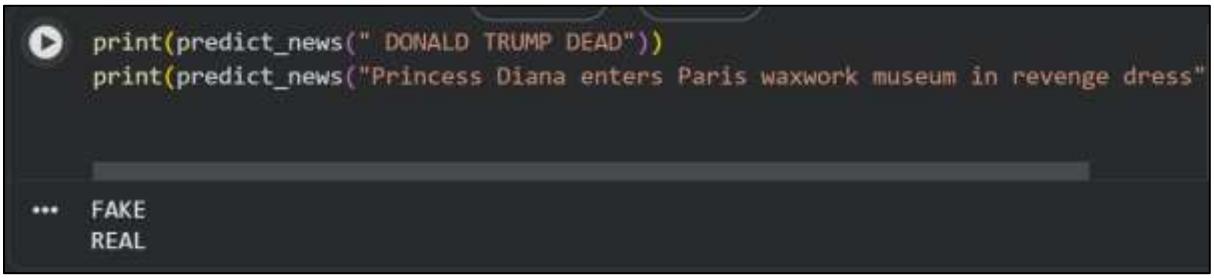
Fake news word cloud



Real news word cloud

8. Sample Predictions

The trained model is tested on custom input sentences to demonstrate real-time prediction capability.



Screenshot showing example predictions such as "FAKE" or "REAL" for given input text.

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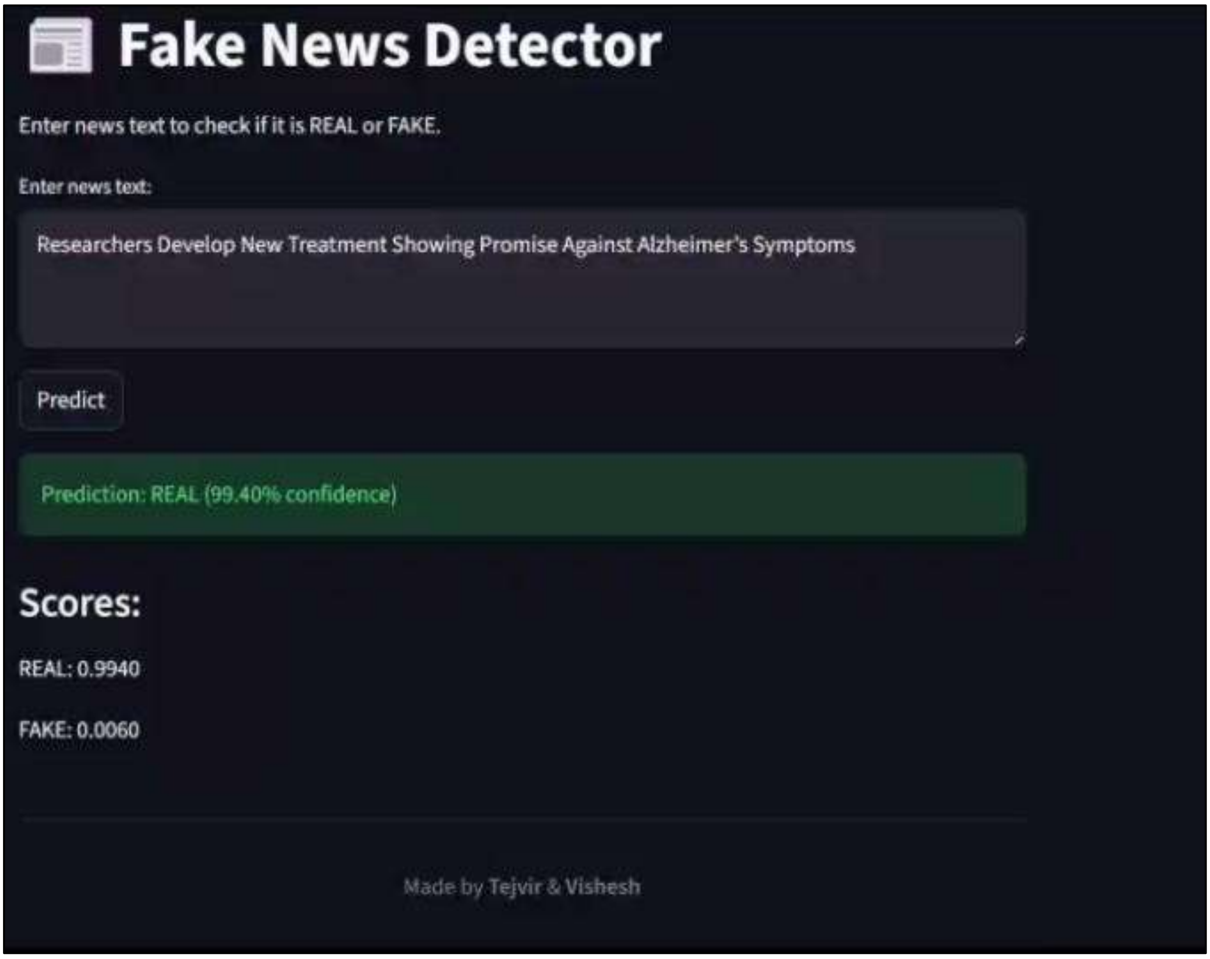
9. Working Prototype Demonstration

A working prototype of the fake news detection system was implemented where a user can input a news headline or article, and the trained BERT model predicts whether the news is Fake or Real in real time.



Screenshot of Prediction: Real News

Screenshot of Prediction: Fake News

Screenshots showing the user input and corresponding model prediction

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10. Conclusion

This project successfully demonstrates the application of a transformer-based deep learning model for fake news detection. The BERT model effectively captures contextual information from news articles, resulting in high classification accuracy and robust performance across multiple evaluation metrics.
The results confirm that modern NLP techniques can play a crucial role in combating misinformation. Future enhancements may include implementing multilingual models, adding source credibility features, and deploying the model as a web application

11. References

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